

Wave Properties

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Amplitude describes wave height from equilibrium, wavelength is the distance between matching points, and frequency is how many cycles pass per unit time; in the same medium, higher frequency corresponds to shorter wavelength.
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Round 1 - Retrieve It

1. What does amplitude measure on a transverse wave?

2. What is wavelength?

3. What is frequency?

4. In the same medium at the same wave speed, how are frequency and wavelength related?

Round 2 - Sketch It

Draw two transverse waves in equal-width boxes: Wave A with larger amplitude and Wave B with shorter wavelength. Label amplitude and wavelength.

Wave Properties - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which wave has greater amplitude: one with 2 cm height from midline or 5 cm?

2. Two waves travel at the same speed. Wave X has higher frequency than Wave Y. Which has shorter wavelength?

3. For sound, what property most directly relates to pitch?

Round 4 - Reason from Evidence

Explain why changing frequency and wavelength in opposite directions is expected when wave speed stays constant.

Wave Properties - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Wave Properties in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect amplitude to energy. How might a larger-amplitude wave differ in effect from a smaller-amplitude wave?

E**Extend**

Create two wave descriptions that have the same frequency but different amplitudes. Predict one observable difference.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Wave Properties

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Maximum displacement from equilibrium; often shown as height from the midline to a crest or trough.
2. Distance between matching points on consecutive waves, such as crest to crest.
3. Number of wave cycles passing a point per unit time.
4. Higher frequency means shorter wavelength; lower frequency means longer wavelength.

Sketch It - look for

Wave A visibly taller from midline; Wave B has more cycles/shorter crest-to-crest spacing across same width; labels placed correctly.

Apply It

1. 5 cm.
2. Wave X.
3. Frequency.

Reason from Evidence - look for

Use relationship $\text{speed} = \text{frequency} \times \text{wavelength}$. If speed stays constant, increasing one factor requires decreasing the other.

ACE - Extend look-for

Same cycle rate/pitch (for sound) but different intensity/loudness or energy; accept other correct wave contexts.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.